

①

Academy DL, NLP, ML
Boot camp
Criss Nails

Module 1 $\Rightarrow$ Python Multi
threading and
multi processing.

① Process $\rightarrow$ Threads:-

multi threading $\rightarrow$ multi
processing.

Q:- What is a Program?

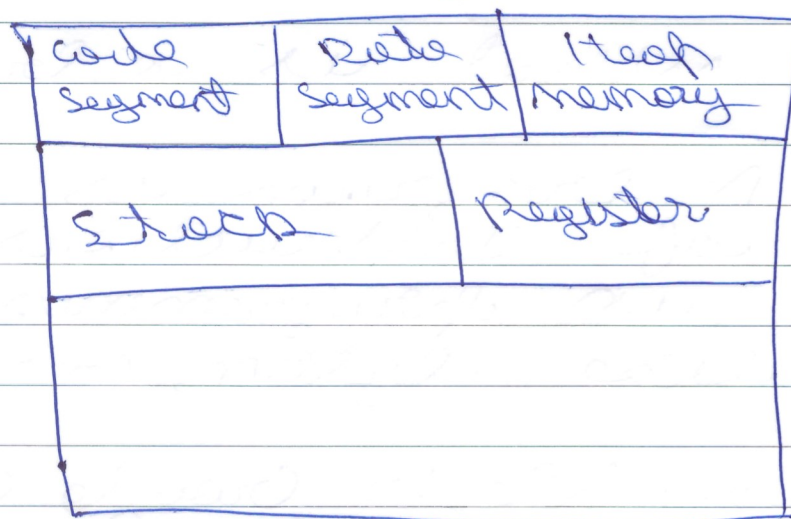
Ans:- A program is a sequence
of instructions written
in programming language
like Python, C++ etc.

Example :- Google Chrome is
an exe file and
it is a type of application
for the program.

Even Word & Excel are also
the type of programs.
P.T.O.

② Process :-

- A process is simply an instance of a program, that is being executed.
- Basically, we're creating an instance of a program that is being executed.
- We need to understand that it works along with the OS or we can say the operating systems.



When we run a process, it will have a separate memory space.

(3)

Since, there is a separate memory space:-

- One process cannot corrupt another. Because, we have a separate memory space.

④ I/O Requirements (Resources):
(Input and Output Requirements)

There will be an increased execution time for switch between the processes

Example:- Browser.

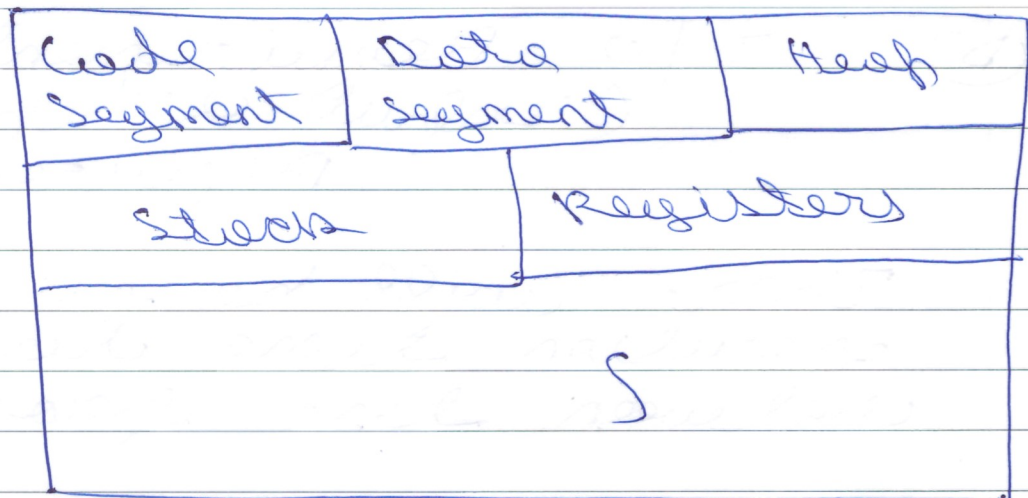
Excel $\rightarrow$ Process $\rightarrow$ I/O

Any installation done for the process will start like an app.

⇒ Threads :-

- A thread is a unit of execution within the process.

• Single Threaded Processes :-



Thread is basically a unit of execution within process.

⇒ Example : we open ms paint and we want to make a rectangle.

- For that, we just go and ~~take a thread~~ a thread is created for execution of process.

⇒ Multi Threaded process

↳ Here everything is same, just there are multiple thread processes taking place.

Code Sequence	Code Segment	Heap
Stack	Region	Registers
	}	}